

Navigating the Mist: A Comprehensive Analysis of Observational Data for Summer Monsoon Precipitation over Indian Subcontinent



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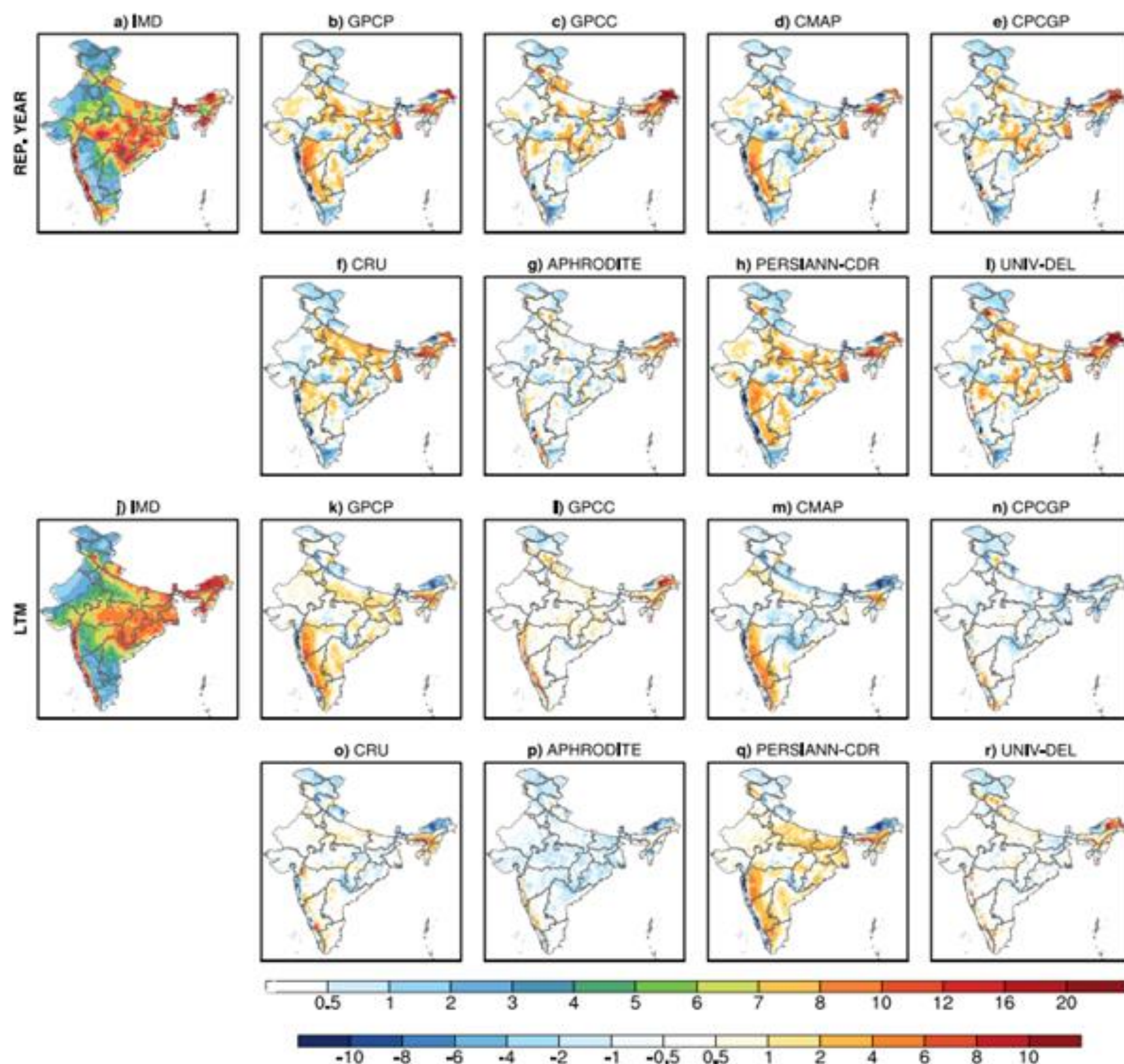
Introduction and Objective

- Precipitation, a vital part of Earth's hydrologic cycle, exerts a profound influence on atmospheric circulation in both weather and climate studies¹. A reliable observation strengthens our understanding of global and regional climate variability and climate change.
- Our path to a perfect observational set is not without hurdles as these datasets vary significantly, and their inherent uncertainties frequently elude users' notice. This leaves us grappling with the pressing question of which dataset aligns best with the specific demands of our research².
- Our aim is to equip researchers with the insights needed to navigate the complex web of precipitation datasets confidently, fostering a more nuanced understanding of regional climate dynamics.

Methodology

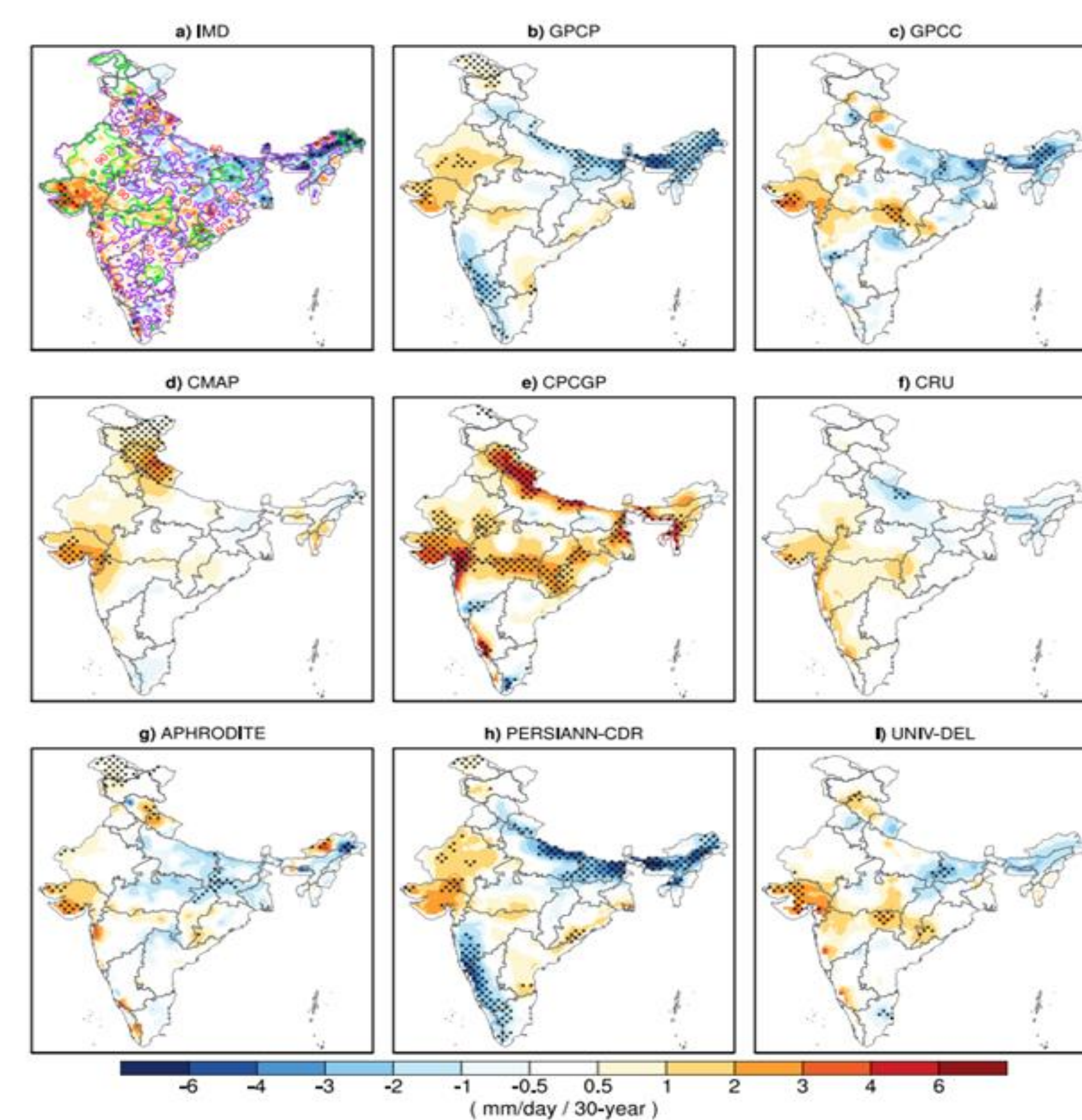
- Ten widely used gridded rainfall datasets in representing Indian summer monsoon rainfall over the Indian landmass are comprehensively evaluated.
- The datasets used are: Gauge-based data (IMD, APHRODITE, GPCC, CRU, UNIVDEL and CPCGP), Satellite data (PERSIANN-CDR and TRMM) and Merged product (CMAP and GPCP).
- Indian Meteorological Department (IMD) gridded rainfall data at 0.25° spatial resolution is used as reference to perform the assessment of all the rainfall products (gauge, satellite and merged) and bias calculations.
- We adopted the Sen's slope as an estimator of the magnitude of trend and the Mann-Kendall (MK) test to assess the significance of trend at a 95% level of confidence.

Results and Conclusions

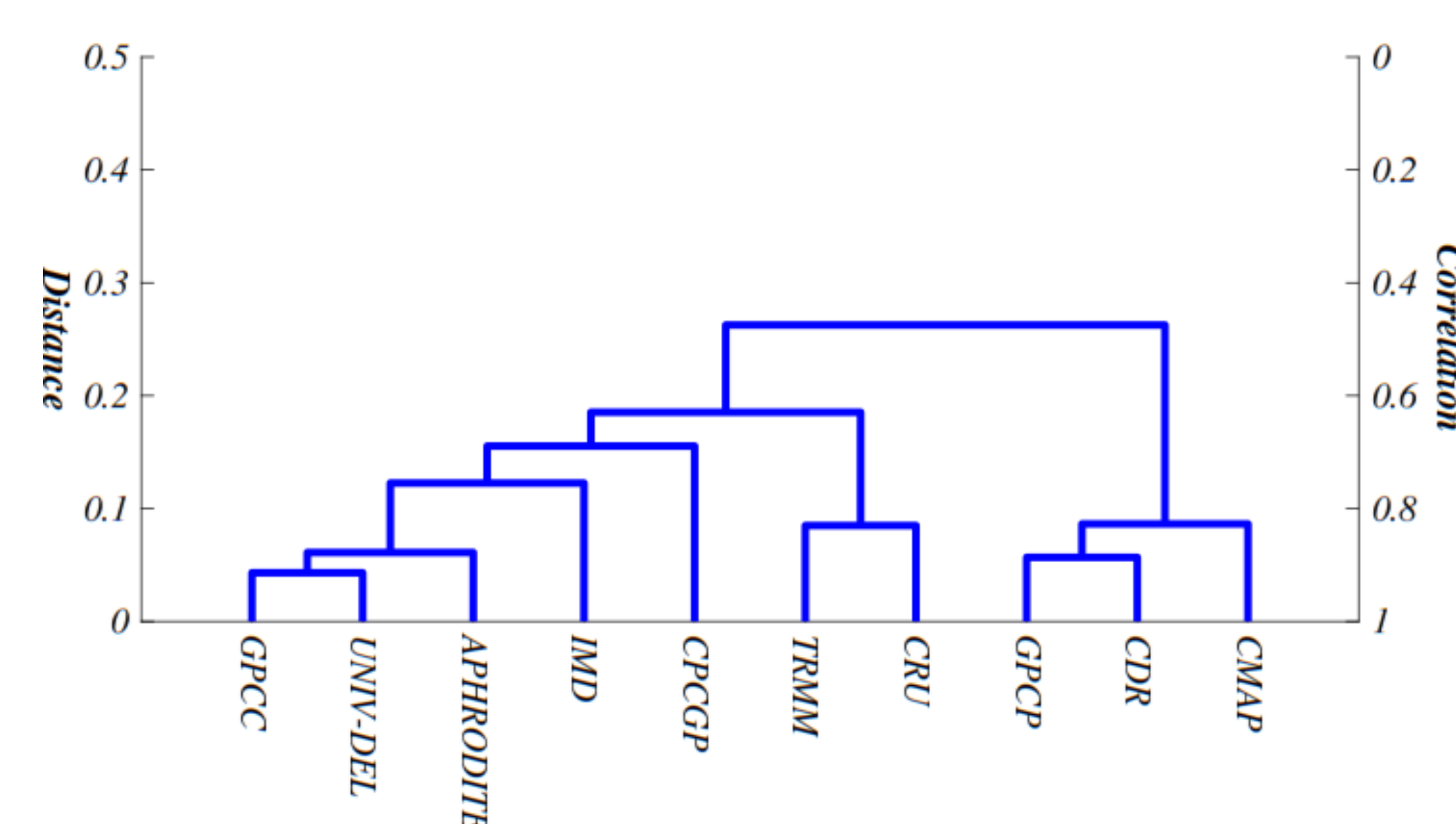


(a) Climatology of IMD precipitation for representative year and anomaly of different datasets with respect to it. (j) climatology of IMD precipitation for long-term mean and anomaly of different datasets with respect to it.

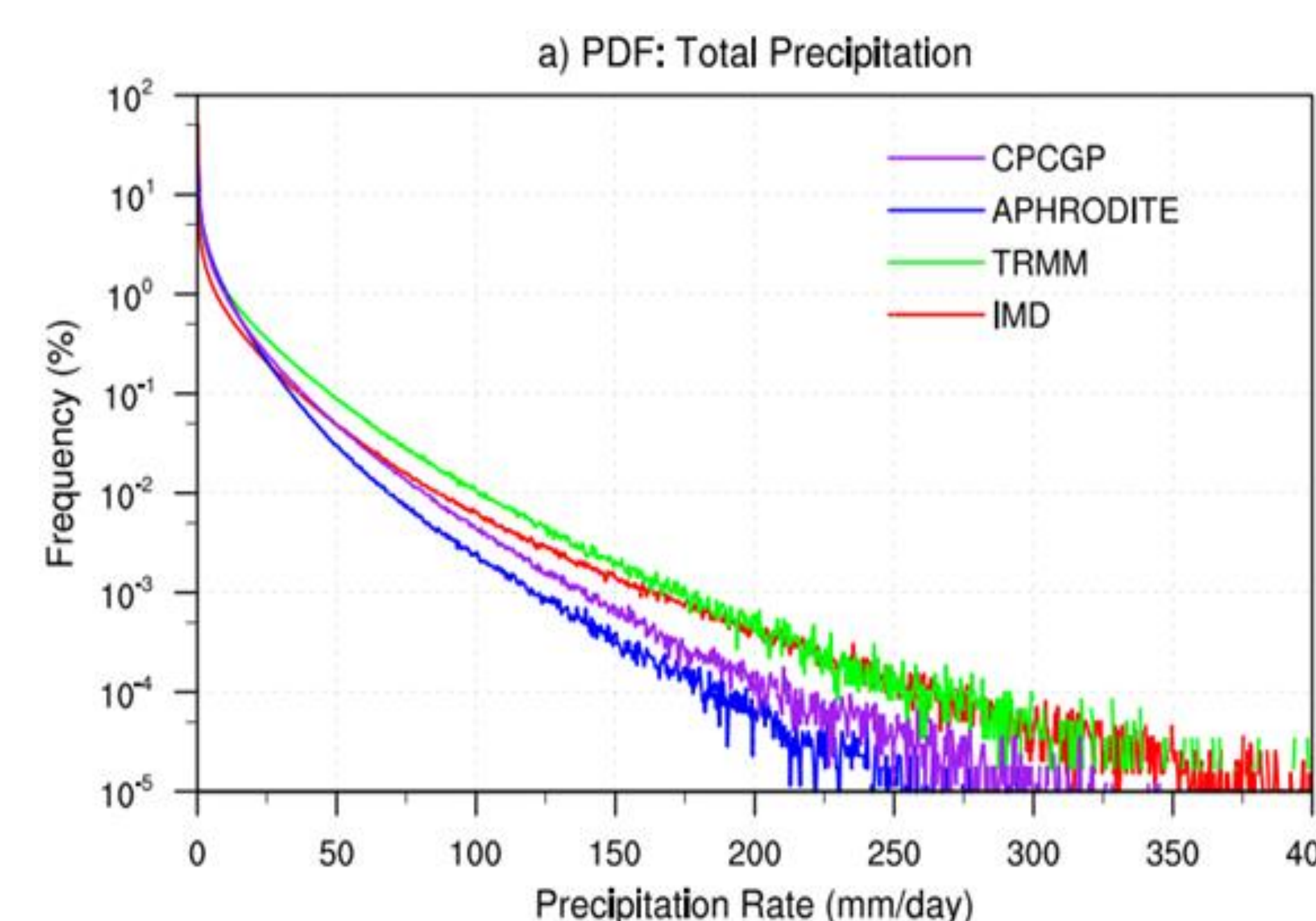
- The uncertainty is varying from -10mm to +10mm in seasonal precipitation in different datasets with respect to IMD long-term mean.
- Gauge-based precipitation estimates are inconsistent between products and vary significantly over the regions having sparse datasets and complex topography.
- The precipitation estimates derived from satellite datasets (PERSIANN-CDR and TRMM) are indirect and have large degree of variability.
- The Western Ghats, Northeastern states and Himalayan states are highly variable region in terms of precipitation.



Linear trend in precipitation in different dataset in 30 years.



Hierarchical clustering in all observed dataset based on the correlation in all India seasonal precipitation.



Probability distribution function analysis shows that at the daily scale, all the datasets have agreement for the light precipitation events.

References

- [1] Eltahir & Bras, 1996. Precipitation recycling.
- [2] Yin et al. 2015. Multi-dataset comparison of gridded observed temperature and precipitation extremes over China.

Acknowledgement

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